

A6-R.R

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```
BASE <- getwd()

DATASET_DIR <- paste(BASE, "/datasets", sep = "")
DATASET_FILE <- "NSS068.csv"
DATASET_PATH <- paste(DATASET_DIR, "/", DATASET_FILE, sep = "")

df <- read.csv(DATASET_PATH)

head(df$foodtotal_v)

## [1] 1141.4924 1244.5535 1050.3154 1142.5917 945.2495 1579.2350

sum(df$state_1 == "MEG")

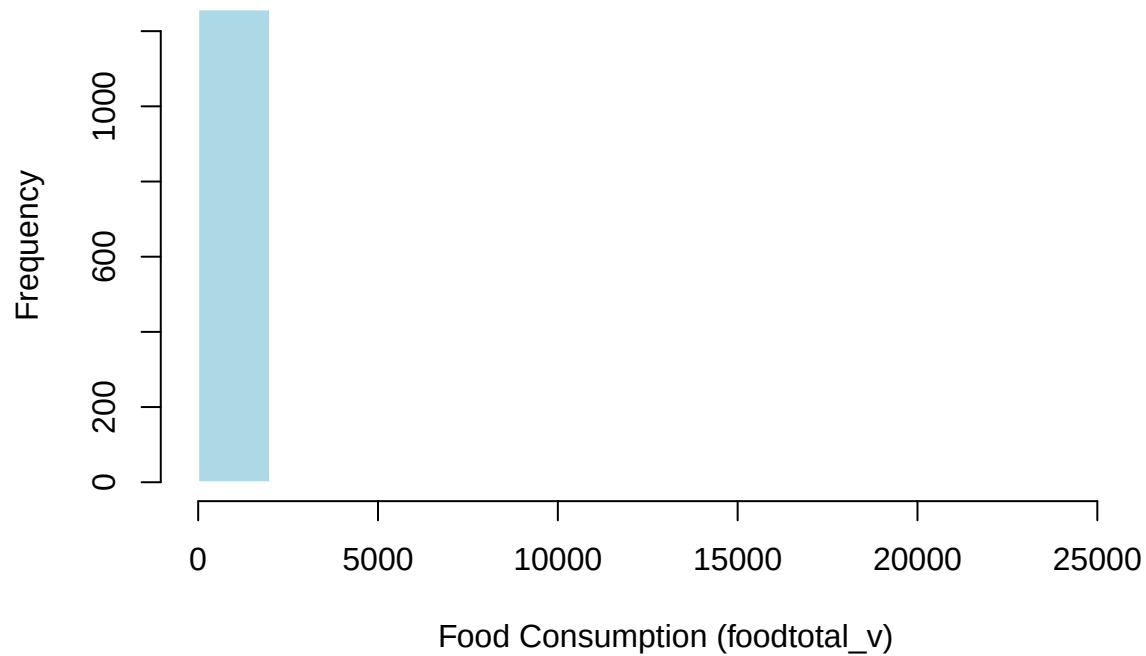
## [1] 1259

meg_df <- df[df$state_1 == "MEG", ]
dim(meg_df)

## [1] 1259 384

hist(
  meg_df$foodtotal_v,
  main = "Distribution of Food Consumption in Meghalaya",
  xlab = "Food Consumption (foodtotal_v)",
  col = "lightblue",
  border = "white"
)
```

Distribution of Food Consumption in Meghalaya



```
meg_df$District <- as.factor(meg_df$District)
```

```
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
meg_df <- meg_df %>%  
  group_by(District) %>%  
  mutate(DWCons = mean(foodtotal_v, na.rm = TRUE)) %>%  
  ungroup()
```

```
district_map <- data.frame(  
  DistrictCode = sprintf("%02d", 1:7),  
  DistrictName = c(  
    "West Garo Hills",  
    "East Garo Hills",  
    "South Garo Hills",  
    "West Khasi Hills",  
    "Ribhoi",
```

```

    "East Khasi Hills",
    "Jaintia Hills"
  ),
  stringsAsFactors = FALSE
)

meg_df <- meg_df %>%
  mutate(DistrictCode = sprintf("%02d", as.numeric(District)))

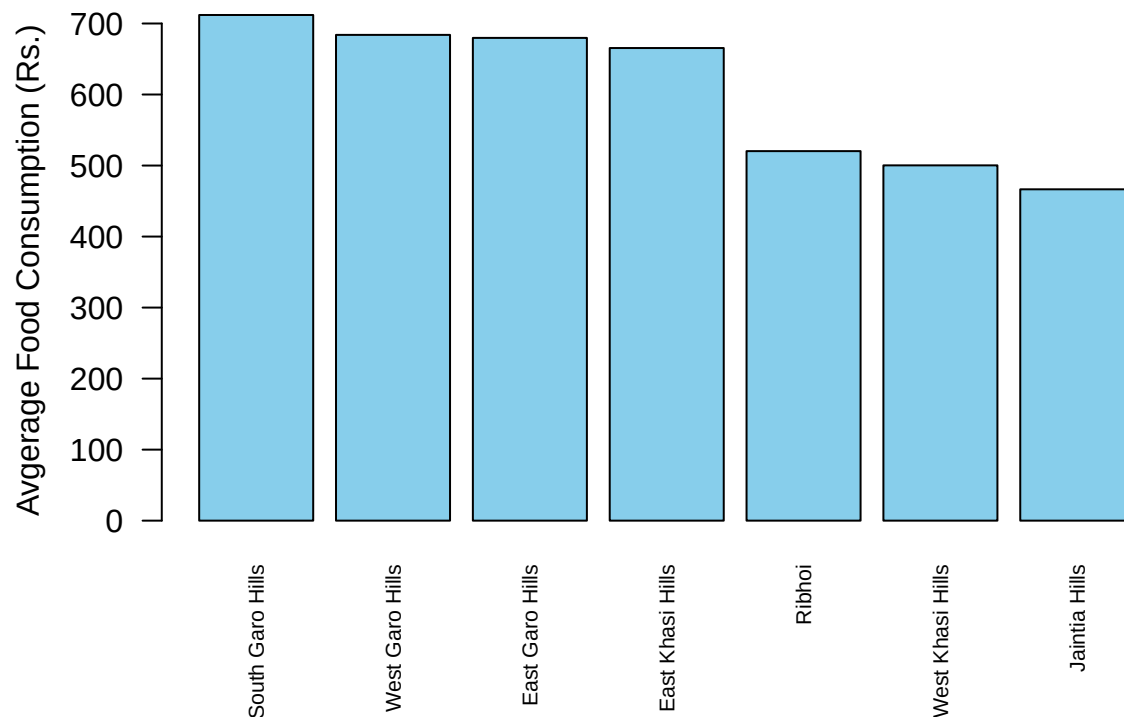
meg_df <- meg_df %>%
  left_join(district_map, by = "DistrictCode")

district_avg <- meg_df %>%
  group_by(DistrictName) %>%
  summarise(avg_food = mean(foodtotal_v, na.rm = TRUE)) %>%
  arrange(desc(avg_food))

barplot(
  height = district_avg$avg_food,
  names.arg = district_avg$DistrictName,
  las = 2,
  col = "skyblue",
  main = "Average Food Consumption by District (Meghalaya)",
  ylab = "Average Food Consumption (Rs.)",
  cex.names = 0.7
)

```

Average Food Consumption by District (Meghalaya)



```
library(ggplot2)
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.10.1, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(dplyr)
```

```
GEOJSON_FILE <- "MEGHALAYA_DISTRICTS.geojson"
GEOJSON_PATH <- paste(DATASET_DIR, "/", GEOJSON_FILE, sep = "")
```

```
data_map <- st_read(GEOJSON_PATH)
```

```
## Reading layer 'MEGHALAYA_DISTRICTS' from data source
##   'C:\Users\ujjwa\Documents\VCU\Pre-Course\SCMA632\Assignments\A6\R\datasets\MEGHALAYA_DISTRICTS.geo.
##   using driver 'GeoJSON'
## Simple feature collection with 11 features and 11 fields
## Geometry type: POLYGON
## Dimension:      XY
## Bounding box:   xmin: 89.81402 ymin: 25.02915 xmax: 92.80282 ymax: 26.1195
## Geodetic CRS:   WGS 84
```

```
data_map <- data_map %>%
  rename(DistrictName = dtname)
```

```
data_map_data <- data_map %>%
  left_join(district_avg, by = "DistrictName")
```

```
data_map_data$avg_food[is.na(data_map_data$avg_food)] <- 0
```

```
ggplot(data_map_data) +
  geom_sf(aes(fill = avg_food, geometry = geometry)) +
  scale_fill_gradient(low = "yellow", high = "red") +
  ggtitle("Average Food Consumption by District (Meghalaya)") +
  theme_minimal() +
  geom_sf_text(aes(label = paste(DistrictName, "-", round(avg_food, 2))), size = 3, color = "black")
```

```
## Warning in st_point_on_surface.sfc(sf::st_zm(x)): st_point_on_surface may not
## give correct results for longitude/latitude data
```

Average Food Consumption by District (Meghalaya)

